Problem Set 5 Solutions

QTM 220

Due 3 November at 2:00pm ET on Canvas. Questions about submission requirements should be directed to your TA.

Gerber, Green, and Larimer Michigan Voter Experiment

Prior to the August 2006 primary election in Michigan, political scientists Gerber, Green, and Larimer conducted a large field experiment in which different mailers were sent to registered voters across Michigan. Because whether or not someone votes (though not who they vote for) is public record, the researchers then accessed the public voting information to measure their outcome– whether an individual voted in the August 2006 primary. The different treatments are as follows:

- Civic Duty: do your civic duty and vote!
- Hawthorne: you are being studied + do your civic duty and vote!
- Self: who votes is public information, here is your recent voting record + do your civic duty and vote!
- Neighbors: what if your neighbors knew whether you voted? here is your recent voting record + your neighbors' recent voting record + do your civic duty and vote!
- Control: no mailer

Households were randomized to receive a mailer (or not) and which type, but voting records are by individual.

The attached data from their experiment has the following variables:

- hh_id : the household id number
- hh_size : number of adults in household
- sex : individual's sex according to voter record

- yob: individual's year of birth
- g2000 : whether or not the individual voted in the general election in 2000
- g2002 : whether or not the individual voted in the general election in 2002
- g2004 : whether or not the individual voted in the general election in 2004
- p2000 : whether or not the individual voted in the primary election in 2000
- p2002 : whether or not the individual voted in the primary election in 2002
- p2004 : whether or not the individual voted in the primary election in 2004
- treatment : the mailer that the household received (or Control for no mailer)
- voted : whether the individual voted in the 2006 primary (outcome)

(a) Create a balance table showing the similarity of individuals assigned to treated and control across sex, age, household size, and turnout in previous elections.

(b) Create a balance table showing the similarity of individuals assigned to each level of treatment and control across sex, age, household size, and turnout in previous elections. Be sure to note how many individuals are in each level of treatment.

(c) Calculate the subsample proportion of 2006 primary turnout within each level of treatment (and control). Plot?

(d) Let's define the following treatment effects of interest: $\hat{\tau}(civic) = \hat{\mu}(civic) - \hat{\mu}(control)$, $\hat{\tau}(hawthorne) = \hat{\mu}(hawthorne) - \hat{\mu}(civic)$, $\hat{\tau}(self) = \hat{\mu}(self) - \hat{\mu}(hawthorne)$, $\hat{\tau}(neighbor) = \hat{\mu}(neighbor) - \hat{\mu}(self)$. Calculate each of these and interpret each in plain language.

(e) Create a bootstrap estimated sampling distribution for each of your τs and a 95% confidence interval for each.

(f) Create four dummy variables– one for each treatment– where an observation is coded as 1 if the house hold received that treatment and 0 otherwise (other treatment or control). Run an additive linear regression of 2006 primary vote on the four treatment variables. Report your results in a table and provide plain language interpretations of your estimates.

(g) Run an additive linear regression of 2006 primary vote on the four treatment variables and the pre-treatment turnout variables (i.e. general election voting in 2002 and 2000, primary election voting in 2004, 2002, and 2000). Report your results in a table (one table for both models is fine) and provide plain language interpretations of your estimates.

(h) BONUS?: Why might we be concerned about using individual turnout as the outcome for this experiment?

Women's Health Initiative

describe data, variable names

(a) Balance table on age

(b) Overall estimated treatment effect (difference in proportions) of CEE+MPA on breast cancer

(c) Bootstrap estimated sampling distribution, 95% confidence interval

(d) Estimate treatment effect within each age group

(e) bootstrap estimated sampling distribution for within-group treatment effects, 95% confidence intervals

(f) additive model

(g) interactive model

read + reflect?

Dogs

The Giant Database of Dogs, a crowdsourced dataset organized by Prof. Leanne Powner, includes the following variables:

- Obs – unique observation number for case identification
- Name – dog's name as provided by the owner (excess text removed)
- Gender – {male, female} closed choice
- Fixed – "yes" if spayed or neutered, as appropriate; "no" if not
- Color – respondents selected one from {light brown, dark brown, black, white, reddish, yellow/blond, no primary color, other}. Other was not an open-ended response.
- Heritage – 1 = single breed, 2 = designer/deliberate mix, 3 = mixed/unknown

Use dogs.csv for the following.

(a) Recode gender and fixed into numerical binary variables. Regress dog's weight on gender and fixed. Report your results in a table and interpret your estimates in plain language.

```r
dogs$num.gender<- ifelse(dogs$Gender=="Male", 1, 0)
dogs$num.fixed<- ifelse(dogs$Fixed=="Yes", 1, 0)
```

```r
dogmod<- lm(Weight ~ num.gender + num.fixed, data=dogs)
summary(dogmod)
```

```
Call:
lm(formula = Weight ~ num.gender + num.fixed, data = dogs)

Residuals:
    Min      1Q  Median      3Q     Max 
-43.162 -22.162  -4.543  16.957 130.838 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)   43.024      5.758   7.472 4.18e-13 ***
num.gender     2.619      2.725   0.961    0.337    
num.fixed      1.519      5.714   0.266    0.790    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 28.72 on 447 degrees of freedom
  (4 observations deleted due to missingness)
Multiple R-squared:  0.002151,	Adjusted R-squared:  -0.002313 
F-statistic: 0.4818 on 2 and 447 DF,  p-value: 0.618
```

(b) Create a scatter plot with weight on the y-axis, fixed on the x-axis, and use different colors for male and female dogs (include a legend). add lines?

(c) Regress dog's weight on gender, fixed, and gender*fixed. Report your results in a table (one table for both models is fine) and provide plain language interpretations of your estimates.

```r
dogmod2<- lm(Weight ~ num.gender*num.fixed, data=dogs)
summary(dogmod2)
```

```
Call:
lm(formula = Weight ~ num.gender * num.fixed, data = dogs)

Residuals:
    Min      1Q  Median      3Q     Max 
-43.337 -22.337  -4.402  17.303 130.663 
```

```
Coefficients:
                     Estimate Std. Error t value Pr(>|t|)    
(Intercept)            46.023      8.666   5.311 1.73e-07 ***
num.gender             -2.441     11.258  -0.217    0.828    
num.fixed              -1.621      8.867  -0.183    0.855    
num.gender:num.fixed    5.376     11.603   0.463    0.643    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 28.74 on 446 degrees of freedom
  (4 observations deleted due to missingness)
Multiple R-squared:  0.002631,  Adjusted R-squared:  -0.004078 
F-statistic: 0.3922 on 3 and 446 DF,  p-value: 0.7587
```

(d) other lines to scatter plot?

(e) Write the equations for each or your models using β_i as your parameters. Fill in the following table using with the βs from your models (not numbers, for example $\beta_0 + \beta_1$.

(f) Fill in the table using the estimates from your models.

(g) Calculate sub-sample means and compare???